

Assignment 4 — C Programming Debugging: Errors & Corrections

Each table lists the buggy line, the type of error, an explanation, and the corrected version.

File: 1.c

Erroneous Code	Error Type	Explanation	Corrected Code
<code>int department = 1</code>	Syntax Error	Missing semicolon at end of statement.	<code>int department = 1;</code>
<code>int marks = "90";</code>	Type Mismatch	Assigning a string literal to an int variable.	<code>int marks = 90;</code>
<code>printf("Outer x = %d\n", x);</code>	Logical Error	Inner block's x=20 only exists inside that block; outer x is still 10 — this is correct, but easy to misread as a bug. (No fix needed; included to verify scope understanding.)	<code>// x correctly remains 10 here</code>
<code>if(x = 10)</code>	Logical Error	Assignment operator '=' used instead of comparison '=='; this assigns 10 to x (always true) instead of comparing.	<code>if(x == 10)</code>
<code>printf("%d\n", score);</code>	Undeclared Variable	'score' was never declared before use.	<code>int score = 90; printf("%d\n", score);</code>
<code>int totalStudents = 100;</code>	Shadowing Issue	Local variable shadows the global totalStudents (declared = 500), hiding the global inside main().	Rename local variable, e.g. <code>int localTotal = 100;</code>
<code>sum = add(a,b);</code>	Linker / Undeclared Function Error	Function add() is never declared or defined in this file.	<code>int add(int a,int b){ return a+b; } // and declare prototype above main()</code>
<code>printf("%d\n", temp);</code>	Scope Error	'temp' was declared inside an inner block and is out of scope here.	Declare temp in the outer scope before the block, or move the printf inside the block.
<code>arr[10] = 100;</code>	Array Bounds Error	arr[5] only has valid indices 0–4; index 10 is out of bounds.	<code>arr[4] = 100; // use a valid index 0-4</code>
<code>printf("%d\n", salary);</code>	Type Mismatch	salary is a float but %d format specifier is used for int.	<code>printf("%.2f\n", salary);</code>
<code>printf("%d\n", count);</code>	Shadowing Issue	Inner block declares its own 'count' (=100), shadowing the outer count (=15); printing after the block shows the outer value, which can confuse beginners.	Use distinct variable names to avoid shadowing, e.g. <code>int innerCount = 100;</code>
<code>printf("%d\n", result);</code>	Scope Error	'result' is declared inside the for-loop block and does not exist outside it.	Declare 'result' before the loop if it must be used after it.
<code>if(age > 18) ... else printf("Eligible\n");</code>	Logical Error	Both if and else branches print the same message 'Eligible', making the condition meaningless.	<code>else printf("Not Eligible\n");</code>
<code>displayData();</code>	Function Call Error	displayData() expects an int parameter but is called with no arguments.	<code>displayData(value);</code>

Erroneous Code	Error Type	Explanation	Corrected Code
<code>char grade = "A";</code>	Type Mismatch	Double quotes create a string literal, but 'grade' is declared as a single char.	<code>char grade = 'A';</code>
<code>printf("%s\n", number);</code>	Type Mismatch	number is an int but %s (string) format specifier is used.	<code>printf("%d\n", number);</code>
<code>while(z > 0); { z--; }</code>	Logical Error	Semicolon after while() creates an empty infinite loop; the block below is unrelated and executes only once.	<code>while(z > 0) { z--; }</code>
<code>int *ptr = NULL; *ptr = 100;</code>	Runtime Error	Dereferencing a NULL pointer causes a segmentation fault (undefined behaviour / crash).	<code>int val = 100; int *ptr = &val; *ptr = 100;</code>
<code>average = calculateAverage(10,20);</code>	Linker Error	Function calculateAverage() is called but never declared/defined.	<code>float calculateAverage(int a,int b){ re turn (a+b)/2.0; }</code>

File: 2.c

Erroneous Code	Error Type	Explanation	Corrected Code
<code>int marks = "90";</code>	Type Mismatch	String literal assigned to an int variable.	<code>int marks = 90;</code>
<code>printf("Program Started\n")</code>	Syntax Error	Missing semicolon at the end of the statement.	<code>printf("Program Started\n");</code>
<code>printf("%d\n", total);</code>	Linker Error	'total' is declared extern but never defined in any file, causing an unresolved external symbol at link time.	<code>int total = 0; // define it, or link the file that defines it</code>
<code>int result = add(10,20);</code>	Linker Error	<code>add()</code> is used before being declared (prototype missing above <code>main</code>); function is only defined later in the file.	Add <code>'int add(int,int);'</code> prototype before <code>main()</code> , or move <code>main()</code> after <code>add()</code> .
<code>printf("%d\n", &number);</code>	Type Mismatch	<code>&number;</code> is an address (pointer) but <code>%d</code> expects an int value.	<code>printf("%d\n", number); // or %p with &number</code>
<code>static int salary = "50000";</code>	Type Mismatch	String literal assigned to int variable.	<code>static int salary = 50000;</code>
<code>printf("%d\n", department);</code>	Scope Error	'department' was declared inside an inner block; it is out of scope at this print statement.	Declare <code>department</code> in the outer scope, or print it inside the block.
<code>printf("%d\n", temp);</code>	Scope Error	'temp' is local to the for-loop block; using it after the loop is out of scope.	Declare <code>temp</code> before the for-loop if needed afterward.
<code>arr[10] = 500;</code>	Array Bounds Error	<code>arr[5]</code> has valid indices 0–4 only.	<code>arr[4] = 500;</code>
<code>printf("%d\n", a/b);</code>	Runtime Error	Division by zero (<code>b = 0</code>) causes undefined behaviour / crash.	<code>if (b != 0) printf("%d\n", a/b); else printf("Cannot divide by zero\n");</code>
<code>if(value = 100)</code>	Logical Error	Assignment <code>'='</code> used instead of comparison <code>'=='</code> .	<code>if(value == 100)</code>
<code>else printf("Pass\n");</code>	Logical Error	Both if and else print 'Pass', so the condition has no real effect.	<code>else printf("Fail\n");</code>
<code>char grade = "A";</code>	Type Mismatch	Double-quoted string assigned to a char; should use single quotes.	<code>char grade = 'A';</code>
<code>printf("%d\n", data);</code>	Logical Error (Uninitialized Variable)	'data' is declared but never initialized before use — prints garbage value.	<code>int data = 0; // initialize before use</code>
<code>int *ptr = NULL; *ptr = 100;</code>	Runtime Error	Dereferencing a NULL pointer; causes a crash.	<code>int val = 100; int *ptr = &val; *ptr = 100;</code>
<code>extern int companyCode; printf("%d\n", companyCode);</code>	Linker Error	<code>companyCode</code> is declared extern but defined nowhere — unresolved symbol.	<code>int companyCode = 0; // define somewhere</code>
<code>average = calculateAverage(10,20);</code>	Linker Error	<code>calculateAverage()</code> is never declared or defined.	<code>float calculateAverage(int a,int b){ return (a+b)/2.0; }</code>

Erroneous Code	Error Type	Explanation	Corrected Code
<code>int employeeId = "EMP101";</code>	Type Mismatch	A string literal cannot be assigned to an int.	<code>char employeeId[10] = "EMP101";</code>
<code>while(counter > 0); { counter--; }</code>	Logical Error	Semicolon creates an empty infinite loop; block below runs only once and is unrelated.	<code>while(counter > 0) { counter--; }</code>

File: 3.c — Student Management System

Erroneous Code	Error Type	Explanation	Corrected Code
<code>int choice</code>	Syntax Error	Missing semicolon after declaration.	<code>int choice;</code>
<code>default printf("Invalid Choice\n");</code>	Syntax Error	'default' label in switch is missing its colon ':'.	<code>default: printf("Invalid Choice\n");</code>
<code>if(index > 5)</code>	Logical / Array Bounds Error	students[] array has size 5 (valid indices 0-4); check should trigger at index ≥ 5 , not > 5 , otherwise index 5 gets written first (out-of-bounds) before the check fires.	<code>if(index $\geq$ 5) { printf("Database Full\n"); } // and check this BEFORE writing to students[index]</code>
<code>for(i=0;i<=totalStudents;i++)</code>	Array Bounds Error	Using $\leq$ causes the loop to access students[totalStudents], one index beyond the last valid student.	<code>for(i=0;i<totalStudents;i++)</code>
<code>return total / 0;</code>	Runtime Error (Logical Error)	Dividing by 0 instead of by totalStudents — should compute the average, not divide by zero.	<code>return total / totalStudents;</code>
<code>int highest = students[0].marks; int lowest = students[0].marks;</code>	Type Mismatch	marks is a float but highest/lowest are declared as int, causing truncation.	<code>float highest = students[0].marks; float lowest = students[0].marks;</code>
<code>printf("Total Students = %d\n", totalStudent);</code>	Undeclared Variable	Variable name misspelled — should be 'totalStudents', not 'totalStudent'.	<code>printf("Total Students = %d\n", totalStudents);</code>
<code>printGrade(); calculateRank();</code>	Linker Error	Functions printGrade() and calculateRank() are called but never declared or defined.	Define these functions, or remove the calls if not needed.
<code>s.rollNo = "101";</code>	Type Mismatch	rollNo is an int field but a string literal is assigned.	<code>s.rollNo = 101;</code>
<code>s.marks = "90";</code>	Type Mismatch	marks is a float field but a string literal is assigned.	<code>s.marks = 90.0;</code>
<code>students[roll] = NULL;</code>	Type Mismatch	students[roll] is a struct, not a pointer — it cannot be assigned NULL.	<code>memset(&students[roll], 0, sizeof(struct Student)); // or use a separate 'isActive' flag</code>
<code>if(name == students[0].name)</code>	Logical Error	Comparing char arrays with $==$ compares pointer addresses, not string contents.	<code>if(strcmp(name, students[0].name) == 0)</code>
<code>fp = fopen("student.txt", "r"); fprintf(fp, ...)</code>	Runtime / Logical Error	File opened in read-only mode ("r") but the code tries to write with fprintf().	<code>fp = fopen("student.txt", "w");</code>
<code>fp = fopen("missing.txt", "r"); fscanf(fp, ...)</code>	Runtime Error	No NULL check after fopen(); if the file doesn't exist, fp is NULL and fscanf() crashes.	<code>if(fp == NULL){ printf("File not found\n"); return; }</code>

File: 4.c — Banking Management System

Erroneous Code	Error Type	Explanation	Corrected Code
<code>if(totalAccounts > 10)</code>	Array Bounds Error	<code>accounts[]</code> has size 10 (valid indices 0-9). Check should be <code>>= 10</code> and should occur BEFORE writing to <code>accounts[totalAccounts]</code> , which can already be out of bounds when <code>totalAccounts==10</code> .	<code>if(totalAccounts >= 10)</code> <code>{ printf("Database Full\n"); return; }</code> <code>// place this check before writing</code>
<code>for(i=0;i<=totalAccounts;i++)</code>	Array Bounds Error	Loop should stop before <code>totalAccounts</code> , not include it, to avoid reading an uninitialized/out-of-range element.	<code>for(i=0;i<totalAccounts;i++)</code>
<code>if(accounts[acc].balance > amount)</code> <code>{ printf("Insufficient Balance\n"); }</code> <code>else { balance - amount; }</code>	Logical Error	Condition is inverted: it prints 'Insufficient Balance' when the balance IS enough, and withdraws money when it is NOT enough.	<code>if(accounts[acc].balance < amount)</code> <code>{ printf("Insufficient Balance\n"); }</code> <code>else { accounts[acc].balance -= amount; }</code>
<code>printf("Total Accounts = %d\n", totalAc count);</code>	Undeclared Variable	Misspelled variable name — should be 'totalAccounts'.	<code>printf("Total Accounts = %d\n", totalAc counts);</code>
<code>calculateInterest();</code> <code>saveAccounts();</code>	Linker Error	<code>calculateInterest()</code> is called but never declared/defined (only 'interestCalculator()' exists, a different name).	<code>Call interestCalculator();</code> instead, or define <code>calculateInterest()</code> .
<code>a.accNo = "1001";</code>	Type Mismatch	<code>accNo</code> is an int field; a string literal cannot be assigned directly.	<code>a.accNo = 1001;</code>
<code>a.balance = "50000";</code>	Type Mismatch	<code>balance</code> is a float field; a string literal cannot be assigned directly.	<code>a.balance = 50000.0;</code>
<code>accounts[accNo] = NULL;</code>	Type Mismatch	<code>accounts[accNo]</code> is a struct, not a pointer, so it cannot hold NULL.	<code>memset(&accounts[accNo], 0, sizeof(stru ct Account));</code>
<code>if(name == accounts[0].name)</code>	Logical Error	Comparing character arrays with <code>==</code> compares addresses, not contents.	<code>if(strcmp(name, accounts[0].name) == 0)</code>
<code>average = generateAverage();</code>	Linker Error	<code>generateAverage()</code> is called but never declared/defined anywhere.	Define <code>generateAverage()</code> , or call an ex isting function such as <code>calculateAverag e()</code> .
<code>fp = fopen("accounts.txt", "r");</code> <code>fprintf(fp, ...)</code>	Runtime / Logical Error	File opened in read mode but written to with <code>fprintf()</code> .	<code>fp = fopen("accounts.txt", "w");</code>
<code>fp = fopen("missing.txt", "r");</code> <code>fscanf(fp, ...)</code>	Runtime Error	No NULL check after <code>fopen()</code> ; crashes if file is missing.	<code>if(fp == NULL){ printf("File not found\ n"); return; }</code>
<code>arr[10] = 500; (runtimeDemo)</code>	Array Bounds Error	Same as earlier files — <code>arr[5]</code> only has indices 0-4.	<code>arr[4] = 500;</code>
<code>int *ptr = NULL;</code> <code>*ptr = 100; (runtimeDemo)</code>	Runtime Error	NULL pointer dereference causes a crash.	<code>int val=100; int *ptr=&val; *ptr=100;</code>

File: 5.c — Library Management System

Note: file 5 repeats the same bug patterns seen in files 3 and 4 (struct field type mismatches, string comparison with ==, NULL assigned to a struct, missing fopen() NULL-check, and off-by-one array bound checks), applied to a Book Issue/Return system.

Erroneous Code	Error Type	Explanation	Corrected Code
<code>b.bookId = "101";</code>	Type Mismatch	bookId is an int field; a string literal cannot be assigned directly.	<code>b.bookId = 101;</code>
<code>b.price = "500";</code>	Type Mismatch	price is a float field; a string literal cannot be assigned directly.	<code>b.price = 500.0;</code>
<code>if(name == accounts[0].name) [pattern: comparing book title]</code>	Logical Error	Comparing char arrays with == compares addresses, not string contents (same bug pattern as files 3 & 4, applied to book titles).	<code>if(strcmp(title, books[0].title) == 0)</code>
<code>books[id] = NULL; (deleteBook pattern)</code>	Type Mismatch	A struct array element cannot be assigned NULL directly.	<code>memset(&books[id], 0, sizeof(struct Book));</code>
<code>fp = fopen("missing.txt", "r"); fscanf(fp, "%d", &totalBooks);</code>	Runtime Error	No NULL check after fopen(); the file 'missing.txt' likely doesn't exist, causing a crash on fscanf().	<code>if(fp == NULL){ printf("File not found\n"); return; }</code>
<code>if(totalBooks > 20) [array bound check pattern]</code>	Array Bounds Error	books[] array has size 20 (valid indices 0-19); the bound check should use >= and should run before writing to the array (same pattern as files 3 & 4).	<code>if(totalBooks >= 20) { printf("Database Full\n"); return; }</code>
<code>for(i=0;i<=totalBooks;i++) [display loop pattern]</code>	Array Bounds Error	Loop condition should be strictly less-than to avoid accessing one element past the last valid book.	<code>for(i=0;i<totalBooks;i++)</code>